P.5 Functions

Name _____ Period ____ Date _____

Pre-Calculus P.5 · Student sheet

I can…

- decide whether a relation is a function.
- evaluate functions, including piecewise functions.
- find the domain of a function and evaluate a difference quotient.

Vocabulary · fill in the missing word

Function — a _____ that assigns each input exactly one output

Domain — the set of _____

Range — the set of _____

Difference quotient — _____

Part 1 · Function or not?

Sort: {(1, 2), (2, 3), (3, 2)}; {(1, 2), (1, 3)}; student → birthday; birthday → student; $y = x^2$; $x = y^2$

Function	Not a function	Why

Part 2 · Notes and examples (fill in each note, then work the example)

Function: each input has exactly _____ output.

Example 1

Is y a function of x?

a. $x^2 + y = 1$

b. $-x + y^2 = 1$

Example 2

$g(x) = -x^2 + 4x + 1$. Find g(2), g(t), and g(x + 2).

Example 3

$f(x) = x^2 + 1$ if $x < 0$; $x - 1$ if $x \geq 0$. Find f(−1), f(0), and f(1).

Implied domain: leave out x-values that make a denominator _____ or an even root negative.

Example 4

Find each domain.

a. $f(x) = \dfrac{1}{x + 5}$

b. $g(x) = \sqrt{4 - x^2}$

Difference quotient: _____, $h \neq 0$

Example 5

Find $\dfrac{f(x + h) - f(x)}{h}$ for $f(x) = x^2 - 4x + 7$.

Part 3 · Partner practice

Partner 1

$f(x) = 3x^2 - 2x$. Find $f(-1)$ and $f(a + 1)$.

Partner 2

Find the domain of $f(x) = \sqrt{2x - 6}$.